

BALANCING REPRESENTATIONS BY IDENTIFYING AND GENERATING UNDER-REPRESENTED DATA

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ABSTRACT

Self-supervised learning inherits the coverage of its source data. When an unlabeled corpus is long-tailed, the representation becomes dense around frequent modes and thin where coverage is weak. Objectives for long-tailed SSL reweight the images already in the corpus and add no new ones. We introduce BRIDGE, (Balancing Representations by Identifying and Generating Under-represented Data), a label-free feedback loop around the source set. The current encoder scores every source image by its mean k NN distance, and a budgeted acquisition rule selects images from a band of that score distribution where support is thin and the generator still produces faithful variants. A frozen image-to-image diffusion model then adds local variants of the selected anchors, and pretraining continues on the enlarged source set. We formulate acquisition as recoverability-weighted allocation of local support and derive when a finite budget should favor a sparse band that excludes the extreme tail and when reallocation after each repair is optimal under a fixed objective. Across five long-tailed, web-scraped and synthetic source regimes, BRIDGE has the highest seven-task linear-probe and k NN average of the compared methods, and paired experiments extend the gain to ViT-S under contrastive and self-distillation objectives, to low-label fine-tuning and to semantic segmentation. The linear-probe gain is smaller under masked reconstruction. Generated repair matches an oracle that restores the withheld real images, and under a frozen encoder the inserted images shrink the neighborhoods of the selected anchors by 4.7 percent against 2.4 percent at unselected images matched on initial sparsity.

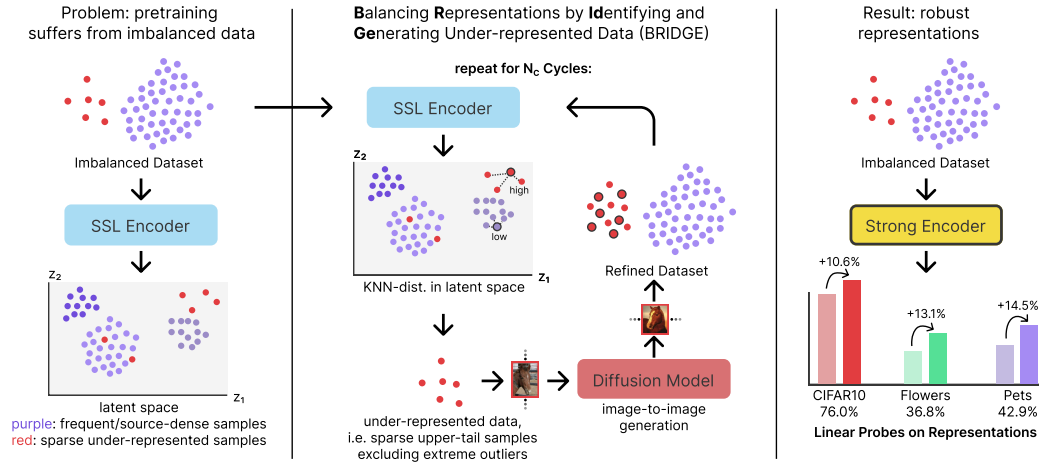


Figure 1: Overview of BRIDGE. Left, an imbalanced source set embedded by the SSL encoder, with sparse under-represented samples in red. Right, the loop. Distance to the k nearest neighbors in the current latent space selects sparse samples from the upper tail of the score distribution while the extreme tail is excluded, image-to-image generation adds variants of them, and the refined dataset is the source set of the next cycle.

1 INTRODUCTION

Self-supervised learning produces transferable visual representations from unlabeled images (Chen et al., 2020; He et al., 2020; Caron et al., 2021; He et al., 2022; Oquab et al., 2024; Siméoni et al., 2025), and every objective learns from what its source corpus contains. When that corpus is long-tailed, the representation concentrates around frequent modes, covers rare regions weakly and transfers worse (Tian et al., 2021; Assran et al., 2023; Kukleva et al., 2023; Jiang et al., 2021). Prior work responds on the model side, with a temperature schedule (Kukleva et al., 2023), a pruned self-competitor (Jiang et al., 2021) or imported external data (Bai et al., 2023), leaving the source set fixed.

We ask whether the source set can be repaired while learning proceeds. **BRIDGE**, **B**alancing **R**epresentations by **I**dentifying and **G**enerating Under-represented Data, embeds the source set with the current encoder after each cycle and scores every image by its mean distance to its k nearest neighbors. It selects a sparse but semantically coherent band of that distribution, generates local variants of the selected images with a frozen image-to-image diffusion model and continues pretraining on the enlarged set (Figure 1). The encoder’s neighbor distances mark the images in thin regions, and the generator produces variants of those images. No labels, class frequencies or downstream signals enter the loop.

A finite generation budget must be placed somewhere in the representation, and the placement may or may not be re-estimated after each repair. We formalize both as budgeted allocation over regions of representation space with diminishing returns to local support and repairability that falls in the extreme tail. The optimal band is then sparse and excludes the extreme tail, greedy reallocation is optimal under a fixed objective, and a rule applied once has an effect bounded, qualitatively, by the ratio of budget to existing support. After one repair the nine score bands lie within seed variation of each other. Over five cycles the three acquisition rules differ by up to eight points.

Our contributions are as follows.

- A theory of label-free data repair as recoverability-weighted allocation of local support (Section 3), in which a repair in a sparse region lowers the error bound more than one in a dense region, greedy reallocation is optimal under a fixed separable objective, and an allocation made from a fixed representation has an effect bounded by the ratio of budget to existing support.
- **BRIDGE**, a label-free feedback loop that instantiates this policy (Section 4) with mean k NN distance as the support estimate, a trimmed score band that holds sparse images the generator can still vary faithfully, farthest-point sampling against redundancy and a frozen image-to-image generator as the repair operator.
- An evaluation on five source regimes and seven transfer tasks under linear-probe and k NN protocols (Section 6), with paired ViT-S experiments across four SSL objectives, low-label fine-tuning, semantic segmentation and an oracle that restores withheld real images.
- An account of where the effect comes from (Sections 6.4 and 6.5). Three acquisition rules separate by eight points over five cycles of repair, and a fixed-encoder intervention test shows the generated support landing at the selected anchors.

2 RELATED WORK

Self-supervised learning under skewed sources. Contrastive, self-distillation and masked-reconstruction pretraining (Chen et al., 2020; He et al., 2020; Caron et al., 2021; Oquab et al., 2024; Siméoni et al., 2025; He et al., 2022) is sensitive to source imbalance, which has been addressed with clustered experts, the hidden uniform cluster prior and reweighting by estimated class frequency (Tian et al., 2021; Assran et al., 2023; Lin et al., 2023). The closest methods change the objective on a fixed corpus, **SDCLR** with a pruned self-competitor (Jiang et al., 2021) and Temperature Schedules by emphasizing head and tail at different times (Kukleva et al., 2023), or import external images (Bai et al., 2023).

Rarity, curation and budgeted acquisition. Mean deep nearest-neighbor distance has been used as a nonparametric out-of-distribution score (Sun et al., 2022), and we use it as a local-support estimate

on the training set itself. Curation by prototype distance or concept-cluster complexity removes examples (Sorscher et al., 2022; Abbas et al., 2024; Whang et al., 2023), SMOTE interpolates labeled minority examples (Chawla et al., 2002), and DOPING oversamples the low-density edge of a learned latent distribution for anomaly detection with no image generator and no recomputed target (Lim et al., 2018), which our top-tail selection rule reproduces under our generator and budget. Active learning spends a finite budget under a learned representation, its core-set form treating acquisition as k -center coverage under the greedy farthest-first traversal we use inside the mode window (Sener and Savarese, 2018; Gonzalez, 1985). ProbCover instead maximizes covered probability mass at a fixed radius (Yehuda et al., 2022), and Hacohen et al. (2022) find dense regions the better target at small budgets.

Generative augmentation and synthetic sources. SDEdit adds noise to an image and denoises it under a diffusion prior (Ho et al., 2020; Rombach et al., 2022; Meng et al., 2022), and that operator is our repair step. It has been applied to few-shot supervised recognition (Trabucco et al., 2024), class-conditional generation improves supervised ImageNet accuracy (Azizi et al., 2023), and synthetic-data utility depends on prompting, filtering, diversity and domain gap (He et al., 2023; Fan et al., 2024). TADA generates faithful variants of the examples a supervised classifier learns slowly (Nguyen et al., 2026), and we adapt its criterion to a label-free learning-speed signal as a baseline. Synthetic imagery has also been a training source for self-supervised features since Ren and Lee (2018), from procedural noise and image programs (Baradad et al., 2021; 2022) to generator latents, synthetic ImageNet clones and text-to-image views (Jahani et al., 2022; Sariyildiz et al., 2023; Tian et al., 2023), none allocating a budget over an observed skewed corpus. DiffAug jointly trains an encoder and a conditional diffusion generator so that generated positives improve contrastive learning (Zang et al., 2024), learning that generator from the small skewed source it is meant to repair, while BRIDGE holds a pretrained generator fixed and asks where it should intervene. Industrial data engines close a loop between a deployed model and its training data, diagnosing failures with vision-language and language models and then curating, auto-labeling and verifying (Liang et al., 2024; Worker and Pathak, 2022). They work from supervised task failures with annotation in the loop and acquire by semantic-cluster occupancy. A frozen generator also brings knowledge from its pretraining data (Li et al., 2023; Tang et al., 2023; Yu et al., 2025), so we report a baseline that distills its features. Appendix A tabulates the closest methods.

3 THEORETICAL FRAMEWORK

Targeted repair is modeled as budgeted allocation over coherent regions of representation space. A repair helps only where it adds support in a region that contributes to downstream error and the generator preserves the content of its anchor, so the policy weighs support deficit against repairability and redundancy. Let $r_k(z; Z)$ be the distance from z to its k th nearest neighbor in Z and A a finite set of valid variants generated from z . Appendix C gives complete proofs.

Proposition 1 (Monotone local support). *For any finite A , $r_k(z; Z \cup A) \leq r_k(z; Z)$. The inequality is strict when an added point enters the first k neighbors of z and displaces a neighbor at strictly larger distance.*

Adding points can only introduce smaller candidate distances, so the k th order statistic cannot increase.

Partition the representation into regions j with effective support n_j and model the regional contribution to downstream error by

$$\mathcal{B}(\mathbf{m}) = \sum_j p_j a_j (n_j + m_j + \kappa)^{-\gamma}, \quad (1)$$

where p_j is the transfer-relevant mass of region j , a_j its difficulty, m_j the number of valid repairs allocated to it, and $\kappa, \gamma > 0$. Write $\Delta_j(m_j) = p_j a_j [(n_j + m_j + \kappa)^{-\gamma} - (n_j + m_j + 1 + \kappa)^{-\gamma}]$ for the reduction from one more valid repair. None of p_j, a_j, κ, γ or the regions is instantiated or estimated in the experiments, and Equation 1 is a qualitative model in the diminishing-returns form of empirical scaling laws (Sorscher et al., 2022).

Proposition 2 (Sparse-region marginal value). *$\Delta_j(m_j)$ is positive and strictly decreasing in $n_j + m_j$. Between two regions of equal transfer mass and difficulty, the region with less effective support has the larger marginal value.*

The forward difference of the convex decreasing map $x \mapsto x^{-\gamma}$ is positive and decreasing, which gives the claim.

Sparsity alone is not sufficient, because generation fails on corrupt or incoherent anchors. Let d be a sparsity score, $b(d)$ the marginal value of a valid repair at that score, $\rho(d)$ the probability of a faithful variant and $\lambda \geq 0$ the cost of an unfaithful one, so the expected repair utility is

$$u(d) = \rho(d) b(d) - (1 - \rho(d)) \lambda. \quad (2)$$

Proposition 3 (Interior recoverable-support optimum). *Suppose b and ρ are positive and continuously differentiable on $[d_{\min}, d_{\max}]$ with $b'(d) > 0$ and $\rho'(d) < 0$, and define $g(d) = b'(d)/(b(d) + \lambda) + \rho'(d)/\rho(d)$. If g is strictly decreasing with $g(d_{\min}) > 0$ and $g(d_{\max}) < 0$, then u has a unique maximizer d^* in the interior, characterized by $g(d^*) = 0$.*

The sign of u' equals the sign of g , so the unique sign change of g is the unique maximum.

The monotonicity assumption on g holds for $b(d) = b_0 d^\beta$ with $\beta \in (0, 1]$ and $\rho(d) = \rho_0 e^{-\mu d}$ (Proposition 6), where g is strictly decreasing for every $\lambda \geq 0$ and $d^* = \beta/\mu$ at $\lambda = 0$. The optimum moves outward when support value is closer to linear and inward when fidelity decays faster. BRIDGE uses a trimmed score band and estimates neither parameter. For the repeated case, consider a separable objective $U(\mathbf{m}) = \sum_j U_j(n_j + m_j)$ with each U_j increasing and discretely concave and a budget of R repairs, $\sum_j m_j = R$.

Proposition 4 (Adaptive allocation under diminishing returns). *For integer allocations, repeatedly assigning one repair to the region with the largest current marginal gain maximizes $U(\mathbf{m})$. A fixed allocation may become suboptimal once a region’s marginal gain falls below that of another region.*

Because every marginal sequence is nonincreasing, the greedy picks are the largest feasible marginals overall. In BRIDGE the encoder changes after every repair, and the regions and their marginals with it, so the benefit of re-estimation in the non-stationary case is an empirical claim.

Proposition 5 (Single-stage insensitivity). *Let $\mathbf{m}$ and $\mathbf{m}'$ be any two allocations of the same budget of R repairs under Equation 1, computed from one representation and applied without re-estimation. Then*

$$|\mathcal{B}(\mathbf{m}) - \mathcal{B}(\mathbf{m}')| \leq R \gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}, \quad (3)$$

which tends to zero as $R/n_{\min} \rightarrow 0$ for the least-supported region $n_{\min}$.

Both allocations subtract at most R marginals from the same baseline, and the mean value theorem bounds every marginal.

Proposition 5 bounds the single-stage gap between any two rules by the ratio of the budget to the support of the least-supported region. The constants are not estimated, so the bound gives no number. With $R = 2,500$ repairs per stage, 500 anchors with five variants each, and the least-supported regions of a 10,000-image source hold far fewer images than that, so the ratio is large and the bound does not constrain the experiments quantitatively. The separation of acquisition rules over five cycles (Section 6.4) lies outside this model, since re-estimation makes each score depend on earlier allocations.

4 BRIDGE: BALANCING REPRESENTATIONS BY IDENTIFYING AND GENERATING UNDER-REPRESENTED DATA

4.1 LOOP AND SCORING

Let $D^{(0)}$ be an unlabeled source set and f_θ an SSL encoder. BRIDGE alternates pretraining and repair over C cycles, with encoder $f_{\theta^{(c)}}$, source set $D^{(c)}$ and embedding set $Z^{(c)}$ at cycle c . The encoder is the only learned component, so the loop applies to any objective that yields embeddings. Every cycle continues from the previous encoder with a fresh optimizer schedule and retains all earlier images.

After training on $D^{(c)}$ we embed each image, $z_i^{(c)} = f_{\theta^{(c)}}(x_i)$, normalize to unit length, and compute

$$d_i^{(c)} = \frac{1}{k} \sum_{z \in \mathcal{N}_k(z_i^{(c)}; Z^{(c)} \setminus \{z_i^{(c)}\})} \|z_i^{(c)} - z\|_2^2, \quad (4)$$

where $\mathcal{N}_k(z; Z)$ is the set of k nearest neighbors of z in Z , so a large score marks an image in a sparse part of the current representation. Scoring uses this squared form. The geometry diagnostic of Section 6.5 uses the unsquared distance to the 20th neighbor under a separate index, a statistic used to measure insertion, with selection based on Equation 4. We use $k = 100$.

4.2 SPARSE AND RECOVERABLE ANCHOR SELECTION

The extreme tail holds artifacts, degenerate crops and images the generator cannot vary faithfully, while uniform sampling spends most of its budget in regions that are already dense. In three steps (Figure 2) the selector restricts attention to the band $U^{(c)} = \{i \mid q_{0.75}^{(c)} \leq d_i^{(c)} \leq q_{0.99}^{(c)}\}$, locates the center $\bar{d}^{(c)}$ of the modal bin of a histogram over that band, forms the pool $P^{(c)} \subseteq U^{(c)}$ of size αB that minimizes $\sum_{i \in P^{(c)}} |d_i^{(c)} - \bar{d}^{(c)}|$, and selects the anchor set $S^{(c)}$ of size B from that pool by greedy farthest-point sampling in embedding space. In every run the modal bin of the right-skewed band sits at its lower edge, so the candidate pool is the $4B$ scores just above the 75th percentile and the histogram step changes nothing (Figure 2). We use $\alpha = 4$ and $B = 500$.

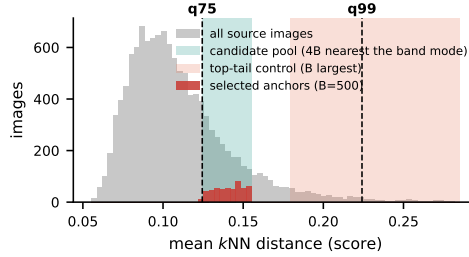


Figure 2: Score histogram of one ImageNet-100-LT run at cycle 0 with the 75th and 99th percentiles marked. The green pool holds the $4B$ scores nearest the modal bin of the band, which lies at its lower edge, the red region is the top-tail selection rule, and the 500 selected anchors are overlaid.

4.3 LOCAL GENERATIVE REPAIR AND CONTINUED PRETRAINING

For each anchor we generate $G = 5$ variants with a frozen image-to-image diffusion model g_ϕ , conditioning on the anchor image under an empty text prompt so the repair signal comes from the anchor alone, and set $D^{(c+1)} = D^{(c)} \cup \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$. The noise ϵ_j indexes independent translations of one anchor, so variants keep coarse structure while changing texture, pose or color. We use Stable Diffusion 3 Medium with 20 denoising steps and strength 0.6 at the training crop resolution, and Algorithm 1 gives the loop.

5 EXPERIMENTAL SETUP

Sources, targets and protocol. We pretrain on five unlabeled source regimes, the long-tailed ImageNet-100-LT (Liu et al., 2019), CIFAR-10-LT and CIFAR-100-LT (Krizhevsky, 2009), and 10,000-image subsets of the web corpora PASS (Asano et al., 2021) and DiffusionDB (Wang et al., 2023). Transfer is measured on seven benchmarks, CIFAR-10-LT, CIFAR-100-LT, Stanford Cars (Krause et al., 2013), FGVC Aircraft (Maji et al., 2013), Oxford Flowers (Nilsback and Zisserman, 2008), Oxford-IIIT Pets (Parkhi et al., 2012) and ImageNet-100-LT, by linear probe and by k NN classification on frozen features. Every number is a three-seed mean and standard deviation, paired comparisons share seeds and checkpoints, and ResNet-50 is the default backbone. The baselines train five cycles of 100 epochs on a source that does not grow, while BRIDGE trains the same epochs on a source that grows from about 10,000 to about 20,000 images, so it receives about 1.5 times the gradient updates in Tables 2 and 3. The cycle ablation and the paired ViT-S experiments match updates exactly. The seed standard deviation of a seven-task average is 0.6 to 1.5 points, so differences below about one point are inside seed variation. Appendix D gives implementation details.

Long-tailed construction. ImageNet-100 is class-balanced, so we construct its long-tailed version. A Pareto profile with exponent $\alpha = 2.5$ over the 100 class ranks is rescaled to per-class retention counts between 1,280 and 5 images and permuted over the classes. Each image is kept with the retention probability of its class. The balanced source of Section 6.1 uses the average of those probabilities for every class, so the two sources have the same expected size and a different class distribution. CIFAR-10-LT and CIFAR-100-LT use their standard configurations.

Table 1: Imposed imbalance on ImageNet-100. A balanced source and a purposefully long-tailed source of the same expected size are compared with BRIDGE on the long-tailed source, under the linear-probe and k NN protocols on seven downstream tasks, mean $\pm$ standard deviation over three seeds. Both sources are pretrained with SimCLR under an identical schedule and differ only in the class distribution of the source. The shaded rows are BRIDGE and bold marks the best value in a column within each protocol. These three runs form one matched comparison and are trained separately from the five source regimes of Table 2.

Protocol	Source	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Linear probe	Balanced	65.3 $\pm$ 1.2	37.0 $\pm$ 0.8	8.0 $\pm$ 0.5	15.1 $\pm$ 0.7	22.4 $\pm$ 1.6	31.2 $\pm$ 2.0	39.0 $\pm$ 1.0	31.1 $\pm$ 1.1
	Purposefully imbalanced	65.4 $\pm$ 4.2	38.6 $\pm$ 1.4	7.6 $\pm$ 1.1	15.3 $\pm$ 0.9	23.7 $\pm$ 1.0	28.4 $\pm$ 1.3	39.6 $\pm$ 0.3	31.2 $\pm$ 1.4
	BRIDGE on the imbalanced source	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
k NN	Balanced	63.6 $\pm$ 0.7	32.8 $\pm$ 0.8	4.4 $\pm$ 0.4	9.0 $\pm$ 0.4	22.3 $\pm$ 0.1	17.2 $\pm$ 0.1	34.0 $\pm$ 1.1	26.2 $\pm$ 0.5
	Purposefully imbalanced	63.9 $\pm$ 0.2	33.2 $\pm$ 0.3	4.5 $\pm$ 0.6	9.0 $\pm$ 0.3	22.5 $\pm$ 0.3	16.8 $\pm$ 0.2	34.3 $\pm$ 0.3	26.3 $\pm$ 0.3
	BRIDGE on the imbalanced source	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5

Table 2: Linear-probe transfer for five source regimes, three label-derived long-tailed sources and two 10k web corpora, on seven downstream tasks, mean $\pm$ standard deviation over three seeds. Every method pretrains on the source in the left column under the same schedule. The shaded row is BRIDGE and bold marks the best value in a column, without resolving overlapping standard deviations. Hybrids of BRIDGE with TS and SDCLR are in Appendix Tables 9 and 10.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.8 $\pm$ 0.2	49.3 $\pm$ 0.2	16.4 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.5 $\pm$ 0.7
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	78.1 $\pm$ 0.3	49.8 $\pm$ 0.4	9.2 $\pm$ 0.8	17.7 $\pm$ 1.0	18.2 $\pm$ 1.1	27.4 $\pm$ 2.7	39.0 $\pm$ 0.6	34.2 $\pm$ 1.0
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	74.2 $\pm$ 0.5	48.1 $\pm$ 0.4	8.3 $\pm$ 1.0	18.2 $\pm$ 0.7	24.4 $\pm$ 2.3	27.6 $\pm$ 0.7	39.8 $\pm$ 1.1	34.4 $\pm$ 1.0
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	73.9 $\pm$ 0.3	47.1 $\pm$ 0.4	13.6 $\pm$ 0.7	20.5 $\pm$ 0.7	37.7 $\pm$ 4.0	37.1 $\pm$ 0.7	49.1 $\pm$ 0.5	39.9 $\pm$ 1.0
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	75.9 $\pm$ 0.4	49.9 $\pm$ 0.5	12.8 $\pm$ 1.0	20.0 $\pm$ 0.4	35.7 $\pm$ 1.1	37.7 $\pm$ 0.8	49.1 $\pm$ 0.5	40.1 $\pm$ 0.7

Compared methods and baselines. The source-regime comparison covers source-matched SimCLR, TS, SDCLR and BRIDGE, with hybrids in the appendix. All ablations use ImageNet-100-LT pretraining, the same downstream suite and three seeds under the five-cycle protocol. The generator ablation includes an oracle that restores 2,500 withheld real images per repair, the same count as the generated variants, placing true support at a deficit no label-free method can see.

6 RESULTS

6.1 IMPOSED IMBALANCE ON IMAGENET-100

ImageNet is class-balanced by construction, so the first question is what a class imbalance costs once it is imposed on purpose. Two sources are drawn from the same pool, one with the power-law profile of Section 5 and one subsampled uniformly at its average retention rate, so they differ only in how images are spread over classes. The two SimCLR runs differ by 0.1 points, 31.1 against 31.2 on the linear-probe seven-task average and 26.2 against 26.3 under k NN. BRIDGE on the imbalanced source reaches 42.3 and 33.6, which is 11.1 and 7.3 points above SimCLR on the same source and a similar margin above the balanced one. Balancing the classes changes the average by 0.1 points, while adding variants in sparse regions raises it by 11.1 points. The remaining experiments use sources that are long-tailed without our intervention.

Table 3: k NN transfer for the five source regimes and methods of Table 2, classified by k NN on frozen features without a learned probe. Mean $\pm$ standard deviation over three seeds, shading and bold as in Table 2.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	7.0 $\pm$ 0.2	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.4
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	74.8 $\pm$ 0.4	41.0 $\pm$ 0.4	4.8 $\pm$ 0.5	10.0 $\pm$ 0.2	24.2 $\pm$ 0.6	16.2 $\pm$ 0.7	29.7 $\pm$ 0.3	28.7 $\pm$ 0.4
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	69.3 $\pm$ 0.5	39.5 $\pm$ 0.1	4.8 $\pm$ 0.6	8.9 $\pm$ 0.1	21.9 $\pm$ 0.1	15.5 $\pm$ 0.1	29.1 $\pm$ 0.7	27.0 $\pm$ 0.3
PASS-10k	SimCLR	67.2 $\pm$ 0.2	37.0 $\pm$ 0.3	5.4 $\pm$ 0.3	10.3 $\pm$ 0.5	30.4 $\pm$ 0.0	16.7 $\pm$ 0.5	38.3 $\pm$ 0.3	29.3 $\pm$ 0.3
	TS	65.2 $\pm$ 0.7	34.6 $\pm$ 0.3	4.4 $\pm$ 0.2	7.9 $\pm$ 0.2	24.5 $\pm$ 0.1	14.4 $\pm$ 0.4	39.3 $\pm$ 0.4	27.2 $\pm$ 0.3
	SDCLR	32.4 $\pm$ 1.5	11.5 $\pm$ 0.8	1.3 $\pm$ 0.2	2.0 $\pm$ 0.1	8.6 $\pm$ 0.7	5.2 $\pm$ 0.2	11.1 $\pm$ 0.8	10.3 $\pm$ 0.6
	BRIDGE (ours)	67.5 $\pm$ 0.3	36.9 $\pm$ 0.3	5.8 $\pm$ 0.5	10.7 $\pm$ 0.4	33.7 $\pm$ 0.2	17.8 $\pm$ 0.6	41.4 $\pm$ 0.8	30.6 $\pm$ 0.4
DiffusionDB-10k	SimCLR	67.3 $\pm$ 0.1	36.2 $\pm$ 0.4	4.3 $\pm$ 0.2	8.8 $\pm$ 0.1	26.8 $\pm$ 0.4	17.3 $\pm$ 0.1	36.7 $\pm$ 0.8	28.2 $\pm$ 0.3
	TS	64.7 $\pm$ 0.4	33.6 $\pm$ 0.3	3.1 $\pm$ 0.5	7.3 $\pm$ 0.3	23.2 $\pm$ 0.4	17.0 $\pm$ 0.0	37.8 $\pm$ 0.6	26.7 $\pm$ 0.4
	SDCLR	32.6 $\pm$ 1.2	10.0 $\pm$ 0.3	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.1 $\pm$ 0.7	4.8 $\pm$ 0.4	10.6 $\pm$ 0.5	9.7 $\pm$ 0.5
	BRIDGE (ours)	68.3 $\pm$ 0.2	37.4 $\pm$ 0.4	5.4 $\pm$ 0.3	9.8 $\pm$ 0.1	30.4 $\pm$ 0.2	18.6 $\pm$ 0.2	39.4 $\pm$ 1.0	29.9 $\pm$ 0.3

Table 4: Paired five-cycle ViT-S experiments on ImageNet-100-LT across four SSL objectives, and end-to-end adaptation of the SimCLR ResNet-50 representations. Both runs of every pair branch from one source checkpoint and receive the same optimizer updates after the branch. Panel A reports per-task linear-probe accuracy with the seven-task linear-probe and k NN averages, mean $\pm$ standard deviation over three seeds, and per-task values are in Appendix Tables 19 and 20. Panel B reports full fine-tuning with 1 and 10 percent of the ImageNet-100-LT labels and PASCAL VOC segmentation, each over three paired seeds. Bold marks the better run of each pair, the shaded row is the run with BRIDGE, and the two average columns are means of the seven task means without a standard deviation.

A. Frozen ViT-S transfer under four objectives										
Objective	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg. linear	Avg. k NN
SimCLR	SSL baseline	60.5 $\pm$ 0.5	35.0 $\pm$ 0.2	7.2 $\pm$ 0.2	10.9 $\pm$ 0.4	32.9 $\pm$ 2.7	25.0 $\pm$ 1.2	42.0 $\pm$ 1.4	30.5	30.4
	with BRIDGE	60.9 $\pm$ 0.3	36.3 $\pm$ 0.1	8.2 $\pm$ 0.4	11.8 $\pm$ 0.4	36.6 $\pm$ 2.9	29.8 $\pm$ 3.0	43.9 $\pm$ 1.1	32.5	31.1
MoCo v3	SSL baseline	60.0 $\pm$ 0.4	34.5 $\pm$ 0.0	7.6 $\pm$ 0.2	11.1 $\pm$ 1.3	30.3 $\pm$ 1.4	29.3 $\pm$ 3.1	44.8 $\pm$ 1.2	31.1	33.0
	with BRIDGE	62.4 $\pm$ 0.5	36.3 $\pm$ 0.1	9.6 $\pm$ 0.3	13.4 $\pm$ 1.0	36.8 $\pm$ 6.1	31.2 $\pm$ 4.0	47.5 $\pm$ 0.9	33.9	32.8
DINO	SSL baseline	58.9 $\pm$ 0.9	33.6 $\pm$ 1.8	8.1 $\pm$ 0.2	11.4 $\pm$ 1.0	38.1 $\pm$ 5.2	27.5 $\pm$ 2.3	44.6 $\pm$ 1.5	31.8	30.5
	with BRIDGE	61.3 $\pm$ 0.1	35.2 $\pm$ 0.3	9.5 $\pm$ 0.9	13.3 $\pm$ 0.6	37.2 $\pm$ 2.6	34.6 $\pm$ 3.6	46.8 $\pm$ 0.8	34.0	32.0
MAE	SSL baseline	37.7 $\pm$ 3.7	13.2 $\pm$ 1.9	1.5 $\pm$ 0.2	2.4 $\pm$ 1.0	10.5 $\pm$ 5.9	10.1 $\pm$ 3.0	19.9 $\pm$ 2.1	13.6	9.6
	with BRIDGE	41.5 $\pm$ 2.2	15.0 $\pm$ 2.4	1.6 $\pm$ 0.4	3.2 $\pm$ 0.7	8.5 $\pm$ 2.3	8.6 $\pm$ 2.2	22.2 $\pm$ 2.4	14.4	11.6
B. End-to-end downstream adaptation, SimCLR ResNet-50										
Evaluation	Setting	SSL baseline			With BRIDGE			Paired difference		
Full fine-tuning	IN100-LT, 1% labels	14.31 $\pm$ 0.33			16.29 $\pm$ 1.11			+1.97 $\pm$ 0.78		
Full fine-tuning	IN100-LT, 10% labels	21.56 $\pm$ 0.75			24.72 $\pm$ 1.75			+3.17 $\pm$ 1.66		
Segmentation	PASCAL VOC, mIoU	12.78 $\pm$ 0.79			15.20 $\pm$ 0.86			+2.42 $\pm$ 0.27		

6.2 TRANSFER ACROSS FIVE SOURCE REGIMES

BRIDGE has the best seven-task average in every source regime under both protocols (Tables 2 and 3), and against SimCLR and TS it is best in 31 of 35 linear-probe cells and 33 of 35 k NN cells. The gain over source-matched SimCLR is above ten linear-probe points under CIFAR-100-LT and above five under CIFAR-10-LT, the two smallest and most skewed sources, and 2.4 to 3.1 points under the three larger sources. TS wins the in-domain target under the ImageNet-100-LT and PASS sources, and SDCLR’s pruning schedule was tuned for larger sources, so at 10,000 images its averages fall to 13 to 15. The DiffusionDB source is itself synthetic and shows the stylistic bias of the generator that produced it, so the added variants there are synthetic images derived from synthetic anchors.

6.3 PRE-TRAINING OBJECTIVES AND END-TO-END ADAPTATION

Four objectives run paired on ViT-S under their reference recipes and one repair budget, and BRIDGE raises the seven-task linear-probe average of each, by 2.0 points for SimCLR, 2.8 for MoCo v3 and 2.2 for DINO, and by 0.8 for MAE, inside seed variation. Under k NN the MoCo v3 average is unchanged to slightly lower, 33.0 against 32.8, while the MAE average rises 2.0 points. Of the 14

Table 5: Ablations on ImageNet-100-LT under the full five-cycle protocol, linear-probe accuracy, mean and standard deviation over three seeds. Each block changes one component, and its shaded row is the default. The selection, cycle and backbone blocks share one default run (42.3) and the generator block’s default is a separate run set (41.5), a gap inside seed variation. Bold marks the best competing setting in a column, excluding the oracle row, which is the oracle of Section 5 restoring the withheld real images. Table 2 reports the same configuration from a separate export, and the two differ by up to 1.2 points on Cars. The k NN counterparts are in Table 6.

Group	Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Generator	Stable Diffusion 3 (default)	75.0 $\pm$ 0.3	48.5 $\pm$ 0.2	16.3 $\pm$ 0.8	20.7 $\pm$ 0.4	35.7 $\pm$ 0.9	41.8 $\pm$ 0.8	52.4 $\pm$ 0.7	41.5 $\pm$ 0.6
	FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	10.3 $\pm$ 1.3	17.5 $\pm$ 0.9	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
	Oracle (withheld real images restored)	74.8 $\pm$ 0.4	47.8 $\pm$ 0.7	13.1 $\pm$ 0.1	20.5 $\pm$ 0.5	40.3 $\pm$ 6.1	38.1 $\pm$ 1.5	50.3 $\pm$ 0.4	40.7 $\pm$ 1.4
Selection	Mode window, lower cutoff p1	75.6 $\pm$ 0.1	49.2 $\pm$ 0.2	16.9 $\pm$ 0.7	21.9 $\pm$ 0.4	34.6 $\pm$ 2.9	43.5 $\pm$ 0.7	52.8 $\pm$ 0.1	42.1 $\pm$ 0.7
	Mode window, lower cutoff p50	75.5 $\pm$ 0.1	49.3 $\pm$ 0.4	15.2 $\pm$ 0.3	22.4 $\pm$ 0.6	35.5 $\pm$ 1.1	43.8 $\pm$ 3.1	53.4 $\pm$ 0.9	42.2 $\pm$ 0.9
	Mode window, lower cutoff p75 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
	Top tail	70.8 $\pm$ 0.5	43.1 $\pm$ 1.0	10.0 $\pm$ 1.7	15.9 $\pm$ 1.3	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4	34.3 $\pm$ 1.2
	Uniform	71.5 $\pm$ 0.2	43.8 $\pm$ 0.7	9.3 $\pm$ 0.9	17.5 $\pm$ 1.2	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9	35.5 $\pm$ 1.2
Cycles	2 cycles	68.9 $\pm$ 0.6	41.6 $\pm$ 1.3	10.6 $\pm$ 1.2	16.4 $\pm$ 0.4	30.7 $\pm$ 4.5	33.2 $\pm$ 0.9	43.3 $\pm$ 0.6	35.0 $\pm$ 1.4
	5 cycles (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
	10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7
Architecture	ResNet-18	68.8 $\pm$ 0.1	41.7 $\pm$ 0.2	13.5 $\pm$ 0.3	19.7 $\pm$ 0.6	38.7 $\pm$ 3.4	38.4 $\pm$ 2.6	50.2 $\pm$ 0.2	38.7 $\pm$ 1.1
	ResNet-50 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
	ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	8.8 $\pm$ 0.5	12.2 $\pm$ 0.2	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
	ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	8.3 $\pm$ 0.5	14.3 $\pm$ 0.8	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

Table 6: k NN counterpart of Table 5. Each block changes one component and holds the rest of the loop fixed, the shaded row is the default of that block’s run set, bold marks the best competing setting in a column, and the oracle row is excluded from the bolding.

Group	Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Generator	Stable Diffusion 3 (default)	68.6 $\pm$ 0.1	37.2 $\pm$ 0.2	6.4 $\pm$ 0.6	11.5 $\pm$ 0.2	35.2 $\pm$ 0.3	24.3 $\pm$ 0.6	46.8 $\pm$ 0.7	32.9 $\pm$ 0.4
	FLUX	65.8 $\pm$ 1.0	35.4 $\pm$ 0.8	5.7 $\pm$ 0.7	9.4 $\pm$ 0.9	27.7 $\pm$ 1.0	18.8 $\pm$ 1.3	38.7 $\pm$ 1.2	28.8 $\pm$ 1.0
	Oracle (withheld real images restored)	68.8 $\pm$ 0.2	37.7 $\pm$ 0.3	6.6 $\pm$ 0.5	11.4 $\pm$ 0.5	32.7 $\pm$ 0.1	23.6 $\pm$ 0.6	44.4 $\pm$ 0.4	32.2 $\pm$ 0.4
Selection	Mode window, lower cutoff p1	69.2 $\pm$ 0.3	37.7 $\pm$ 0.2	6.6 $\pm$ 0.2	11.4 $\pm$ 0.2	35.9 $\pm$ 0.2	25.0 $\pm$ 0.5	48.3 $\pm$ 0.3	33.4 $\pm$ 0.3
	Mode window, lower cutoff p50	68.7 $\pm$ 0.1	38.0 $\pm$ 0.2	7.3 $\pm$ 0.2	11.3 $\pm$ 0.2	36.5 $\pm$ 0.1	24.8 $\pm$ 0.8	48.5 $\pm$ 0.4	33.6 $\pm$ 0.3
	Mode window, lower cutoff p75 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
	Top tail	65.7 $\pm$ 0.8	34.4 $\pm$ 0.2	4.1 $\pm$ 0.4	9.6 $\pm$ 0.2	23.9 $\pm$ 0.9	18.1 $\pm$ 0.8	35.4 $\pm$ 1.2	27.3 $\pm$ 0.6
	Uniform	66.1 $\pm$ 1.0	35.7 $\pm$ 0.3	4.9 $\pm$ 0.1	10.2 $\pm$ 0.7	25.3 $\pm$ 0.4	19.0 $\pm$ 0.8	37.4 $\pm$ 0.6	28.4 $\pm$ 0.6
Cycles	2 cycles	65.3 $\pm$ 0.4	34.2 $\pm$ 0.5	5.0 $\pm$ 0.7	10.0 $\pm$ 0.2	26.1 $\pm$ 0.8	18.7 $\pm$ 0.6	35.9 $\pm$ 0.3	27.9 $\pm$ 0.5
	5 cycles (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
	10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	10.4 $\pm$ 0.6	25.2 $\pm$ 0.8	18.6 $\pm$ 0.9	36.0 $\pm$ 0.7	28.1 $\pm$ 0.6
Architecture	ResNet-18	66.3 $\pm$ 0.2	35.7 $\pm$ 0.3	5.9 $\pm$ 0.2	11.1 $\pm$ 0.4	33.5 $\pm$ 0.1	23.3 $\pm$ 0.5	44.5 $\pm$ 0.7	31.5 $\pm$ 0.4
	ResNet-50 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
	ViT-S	67.3 $\pm$ 0.4	37.5 $\pm$ 0.3	5.3 $\pm$ 0.2	7.6 $\pm$ 0.1	36.2 $\pm$ 0.7	20.1 $\pm$ 1.0	42.1 $\pm$ 1.0	30.9 $\pm$ 0.5
	ViT-B	69.3 $\pm$ 0.8	40.1 $\pm$ 0.8	6.0 $\pm$ 0.2	8.3 $\pm$ 0.2	39.8 $\pm$ 0.8	21.9 $\pm$ 1.4	45.1 $\pm$ 0.7	32.9 $\pm$ 0.7

measurements per objective, 12 or 13 improve, and fine-grained targets gain most, up to 7.1 points on Pets for DINO. The same ResNet-50 representations improve after end-to-end adaptation, by 2.0 and 3.2 points of fine-tuning accuracy with 1 and 10 percent of the labels and by 2.4 mIoU on PASCAL VOC segmentation.

6.4 ABLATIONS

Generator. Generated repair with Stable Diffusion 3 matches the oracle of Section 5, 41.5 against 40.7. With FLUX at six sampling steps and guidance 2.5, the average falls to 35.0, 6.5 points below Stable Diffusion 3.

Selection. The three mode windows with lower cutoffs at the 1st, 50th and 75th percentile reach 42.1 to 42.3 and are not separated, while uniform acquisition reaches 35.5 ± 1.2 and the top tail 34.3 ± 1.2 . Welch t-tests with Holm adjustment place every window ahead of both at adjusted $p \leq .011$, which agrees with Proposition 3, where excluding the extreme tail raises the objective, and leaves its lower boundary unresolved. Each window changes trimming and diversification at once, and the 6.6 to 8.0 points include both effects.

Cycles and backbone. At fixed total repair volume and optimizer budget, five cycles improve the seven-task average over two by 7.3 points and remain 8.2 points ahead of ten. With two cycles, most of the budget is spent before the representation has changed. With ten cycles, each repair is followed

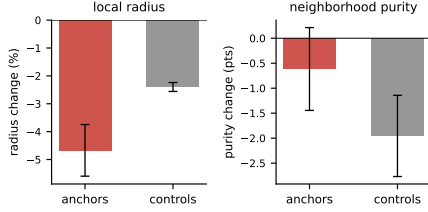


Figure 3: Fixed-encoder intervention test. With the cycle-0 encoder frozen and every generated image in the k NN index, the radius of the treated anchors falls about twice as far as that of matched unselected images, so the variants land in their own anchors’ neighborhoods, and purity falls about a third as much at the anchors. Bars are seed means with one standard deviation.

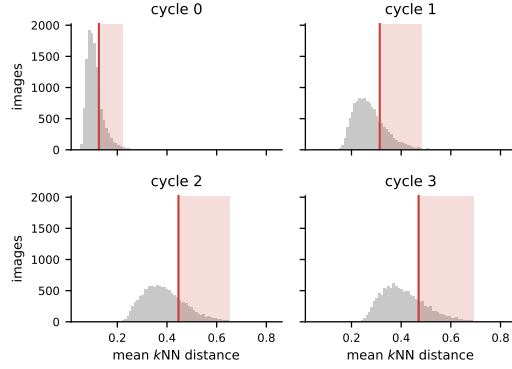


Figure 4: Score distribution over the four repair cycles of one ImageNet-100-LT run, with the 75th to 99th percentile band shaded and its modal bin marked. The distribution shifts to larger scores at every cycle, so the band is re-derived from the current representation.

by only 50 epochs of training. Among the tested backbones, ResNet-50 has the highest average at this source size.

6.5 REPRESENTATION GEOMETRY

Under the frozen cycle-0 encoder (Figure 3), inserting the generated variants reduces the radius of the treated anchors by 4.7 percent against 2.4 percent for unselected images matched within class on initial radius, a paired difference of -2.3 points. Purity falls by 0.6 points at the anchors and by 2.0 at the matched images, so near the anchors most inserted variants share the query’s class, and near the matched images more of them come from other classes. The diagnostic index uses $k = 20$, not the $k = 100$ of selection. Per-seed values are in Appendix B.

7 LIMITATIONS

BRIDGE depends on informative encoder neighborhoods in the first cycle and on a frozen generator that imports the biases of its own pretraining data. Repair fidelity, the ρ of Section 3, is not measured in pixel or semantic space, and the deficit of top-tail selection is consistent with low fidelity there. Later cycles select generated images as anchors, 7 to 13 percent at cycle 1 and 34 to 45 percent at cycle 4, with no filter applied. The band, neighborhood size and budget were held at their defaults across all five sources. The loop adds 30 to 45 percent wall-clock time, mostly diffusion sampling (Appendix E).

8 CONCLUSION

Without labels, BRIDGE uses the encoder’s neighbor distances to select a sparse band of images away from the extreme tail, generates local variants with a frozen diffusion model and retrains, over five cycles. Section 3 motivates the band by diminishing returns to local support and fidelity decay in the extreme tail. Transfer improves on five pretraining sources and under four self-supervised objectives, the loop reaches the accuracy of an oracle that restores withheld real images, the acquisition rule separates from uniform and top-tail acquisition by eight points, and a frozen-encoder test places the generated images in their anchors’ neighborhoods.

Measuring repair fidelity would account for the generator’s biases and could set the band edges in place of fixed percentiles. Other modalities need only an encoder and a generator.

AI USE STATEMENT

In this work we used generative AI tools for writing and debugging the data generators, the training and evaluation scripts and the analysis code, for tabulating results, and for editing the text of the manuscript under the direction of the authors. We also used them to check the bibliography, and every entry was subsequently verified against its publisher, proceedings or arXiv record by the authors. We have not used generative AI tools for generating research ideas. The generated images that enter pretraining come from the frozen image generator described in Section 4, and no assistant-written text or code is part of that data. All AI-assisted code was reviewed and tested by the authors, every number in the paper was checked against the evaluation outputs, and all text was reviewed and revised by the authors. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

Section 4 and Algorithm 1 specify the loop, Section 5 and Appendix D give every operating parameter, schedule, generator setting and data split, and Appendix C contains complete proofs with their assumptions stated in Section 3. Appendix H reports every per-task number behind the main-text averages. An anonymous code release with configuration files for every experiment is linked in Appendix D.

ETHICS STATEMENT

The work uses public datasets under their release terms and a frozen pretrained image generator, releases no new generator, scraped collection or generated corpus, and involves no human subjects. Generative repair inherits the biases and memorization risks of the generator’s pretraining data, and a deployment would require a task-specific review of fairness, privacy and content safety, as Appendix G discusses.

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A COMPARISON WITH RELATED ACQUISITION METHODS

Classical oversampling and TADA receive labels or supervised learning signals (Table 7). DOPING uses latent rarity, targets anomaly detection and does not close the loop through a changing SSL representation. GenRep and StableRep make synthetic images or latent neighborhoods the source of contrastive views without allocating a repair budget over an observed skewed corpus. DiffAug is the prior SSL method most similar to ours, since it also iterates between encoder and generator, and it jointly learns a generator for generic positive views without asking where a frozen generator should intervene. AIDE and industrial data engines retrain a model on data collected where it fails a supervised task, using VLM and LLM reasoning, annotation and domain-specific verification. BRIDGE studies the label-free case in which local support in the current SSL representation is both the diagnostic and the acquisition signal.

Table 7: Closest methods and the dimensions relevant to the acquisition problem. Feedback means that the learner or task model changes the next acquisition decision, and the methods are cited in Section 2.

Method	Labels or task loss	Acquisition signal	Data action	Feedback
SMOTE	class labels	class minority	interpolate examples	no
DOPING	no labels	latent edge density	latent oversampling	no
Concept pruning	no task labels	cluster complexity	remove examples	no
DA-Fusion	labels or few-shot set	class and exemplar	image-conditioned diffusion	no
TADA	supervised loss dynamics	slow-learned examples	faithful diffusion variants	no
GenRep and StableRep	prompts or generator latent	global sampling	synthetic SSL views	no
DiffAug	no labels	learned positive-view objective	jointly learned diffusion views	yes
AIDE	task labels and failures	VLM and LLM failure analysis	curate, generate, annotate	yes
BRIDGE	no labels	local support in current SSL space	budgeted local variants	yes

B REPRESENTATION DIAGNOSTICS

The fixed-encoder intervention test freezes the cycle-0 encoder and queries the same original images before and after inserting all generated samples, so any change reflects the inserted points alone. First-cycle anchors are matched one-to-one within class to original images with the closest initial k NN radius that are never selected in any cycle, and Table 8 reports the per-seed values behind Figure 3. Figure 5 draws every paired ViT-S measurement of Table 4 as a difference. The learned-space comparison retrains the encoder after the intervention and evaluates the same original-image

Table 8: Fixed-encoder intervention test behind Figure 3. The cycle-0 encoder is frozen, every generated image of the run is inserted into the k NN index ($k = 20$), and the same original images are queried before and after. Anchors are the images selected at cycle 0, matched images are paired one-to-one within class on initial radius and are never selected in any cycle. Radius change is relative to the initial radius, purity is the fraction of the 20 neighbors sharing the query label, and the last two columns are the anchor-minus-matched differences.

Seed	Radius change (%)		Purity change (pts)		Difference	
	Anchors	Matched	Anchors	Matched	Radius	Purity
0	-5.38	-2.27	-1.57	-2.79	-3.11	+1.22
1	-5.01	-2.35	-0.20	-1.91	-2.66	+1.70
2	-3.62	-2.58	-0.07	-1.17	-1.05	+1.09
Mean	-4.67	-2.40	-0.61	-1.96	-2.27	+1.34

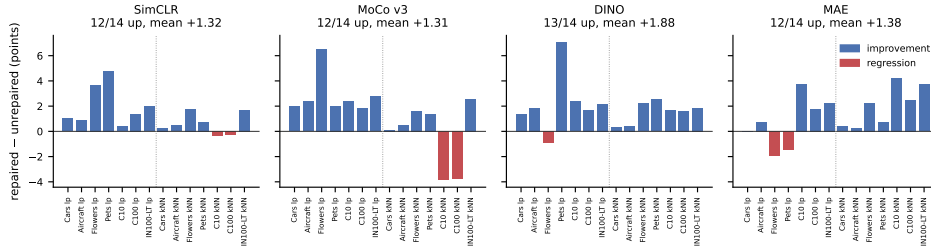


Figure 5: Repair against no repair for four objectives on ViT-S, one bar per downstream measurement (linear probe and k NN on each of seven datasets), from the paired runs of Table 4. Blue bars are improvements. Red bars are regressions, two for SimCLR and MoCo v3 (CIFAR-10 and CIFAR-100 under k NN), one for DINO (Flowers) and two for MAE (Flowers and Pets under the linear probe).

panel. On that panel MoCo v3 on ViT-S gains 21.97 ± 1.58 in effective rank, 0.296 ± 0.019 in spectral entropy and 3.80 ± 0.80 points of 1-NN accuracy after repair, and the first two survive Holm correction, whereas SimCLR improves transfer while its effective rank falls and its local effect is mixed. We therefore use the fixed-encoder result to locate where the variants land in representation space and the learned-space statistics as objective-dependent diagnostics.

Where the temperature-schedule hybrid helps. Combining BRIDGE with TS lowers the seven-task average in every source regime, and it wins 7 of the 70 individual cells (Appendix Tables 9 and 10). Six of those seven fall on CIFAR-10 and ImageNet-100-LT, the two long-tailed downstream targets, and the largest are +3.3 and +2.8 points on ImageNet-100-LT and CIFAR-10 under a PASS and a CIFAR-10-LT source. The fine-grained targets lose in almost every case. TS reweights the contrastive temperature toward rare samples, which sharpens the head-to-tail boundary that the long-tailed targets score, and the hybrid gains on those targets. BRIDGE alone gains more on the four fine-grained targets.

C PROOFS FOR THE SUPPORT-ALLOCATION FRAMEWORK

Proposition 6 (Closed-form interior optimum). *Let $b(d) = b_0 d^\beta$ with $b_0 > 0$ and $\beta \in (0, 1]$, and let $\rho(d) = \rho_0 e^{-\mu d}$ with $\mu > 0$. Then g is strictly decreasing on $(0, \infty)$ for every $\lambda \geq 0$, so Proposition 3 applies whenever g changes sign on the interval. If $\lambda = 0$ the maximizer is $d^* = \beta/\mu$, independent of b_0 and ρ_0 .*

Proof of Proposition 1. Sort the distances from z to points in Z and denote their k th order statistic by $r_k(z; Z)$. Adding A preserves every old candidate distance and may introduce smaller ones. The k th order statistic can therefore only decrease. It decreases strictly when an added distance enters the first k positions and displaces a strictly larger distance. $\square$

Proof of Proposition 2. Let $x = n_j + m_j + \kappa$. Since $f(x) = x^{-\gamma}$ is positive, strictly decreasing and strictly convex for $x > 0$, the forward difference $f(x) - f(x + 1)$ is positive and strictly decreases

with x . Multiplication by $p_j a_j > 0$ preserves both properties. Equal $p_j a_j$ therefore leaves effective support as the sole determinant of the marginal ordering. $\square$

Proof of Proposition 3. Rewrite the utility as $u(d) = \rho(d)(b(d) + \lambda) - \lambda$. Its derivative is

$$u'(d) = \rho(d)(b(d) + \lambda) \left[\frac{b'(d)}{b(d) + \lambda} + \frac{\rho'(d)}{\rho(d)} \right] = \rho(d)(b(d) + \lambda) g(d).$$

The factor preceding $g(d)$ is positive. Since g is strictly decreasing and changes sign across the interval, it has exactly one root d^* by continuity. Hence u increases before d^* and decreases after it, making d^* the unique global maximizer. $\square$

Proof of Proposition 6. With $b(d) = b_0 d^\beta$ and $\rho(d) = \rho_0 e^{-\mu d}$ we have $\rho'(d)/\rho(d) = -\mu$, so

$$g(d) = \frac{b_0 \beta d^{\beta-1}}{b_0 d^\beta + \lambda} - \mu = \frac{\beta d^{\beta-1}}{d^\beta + c} - \mu, \quad c = \lambda/b_0 \geq 0.$$

Write $h(d) = \beta d^{\beta-1} (d^\beta + c)^{-1}$. Differentiating,

$$h'(d) = \frac{\beta d^{\beta-2} [(\beta-1)(d^\beta + c) - \beta d^\beta]}{(d^\beta + c)^2} = \frac{\beta d^{\beta-2} [(\beta-1)c - d^\beta]}{(d^\beta + c)^2}.$$

For $\beta \in (0, 1]$ we have $(\beta-1)c \leq 0 < d^\beta$ for all $d > 0$, so the bracket is strictly negative and $h'(d) < 0$. Hence $g = h - \mu$ is strictly decreasing for every $\lambda \geq 0$. When $\lambda = 0$ we have $c = 0$ and $g(d) = \beta/d - \mu$, with unique root is $d^* = \beta/\mu$. Since g is positive below it and negative above it, d^* is the maximizer by the argument of Proposition 3. $\square$

Proof of Proposition 4. For region j define the marginal sequence $\delta_{j,t} = U_j(n_j + t) - U_j(n_j + t - 1)$ for $t \geq 1$, which discrete concavity makes nonincreasing, so every feasible allocation of R repairs selects a prefix of length m_j from each sequence and its utility is the baseline plus the selected prefix marginals. After t greedy steps the selected set is the t largest prefix-feasible marginals, and the next greedy pick is the largest remaining prefix-feasible marginal, so the set stays the top $t + 1$ and the claim follows by induction on t . No feasible allocation can therefore exceed the greedy total. For the second sentence take two regions with marginal sequences $(3, 1, 1)$ and $(2, 2, 2)$ and a budget of three, where the fixed allocation $(3, 0)$ built from the initial ordering yields 5 while greedy yields $3 + 2 + 2 = 7$. $\square$

Proof of Proposition 5. Write $f(x) = x^{-\gamma}$. For $x > 0$ the mean value theorem gives $f(x) - f(x + 1) = \gamma \xi^{-\gamma-1}$ for some $\xi \in (x, x + 1)$, so

$$0 < \Delta_j(m_j) = p_j a_j [f(n_j + m_j + \kappa) - f(n_j + m_j + 1 + \kappa)] \leq p_j a_j \gamma (n_j + \kappa)^{-\gamma-1},$$

using $\xi > n_j + m_j + \kappa \geq n_j + \kappa$ and $m_j \geq 0$. Both $\mathcal{B}(\mathbf{m})$ and $\mathcal{B}(\mathbf{m}')$ are obtained from the common baseline $\mathcal{B}(\mathbf{0})$ by subtracting sums of at most R such marginals, since $\sum_j m_j = \sum_j m'_j = R$. Each sum lies in $[0, R\gamma \max_j p_j a_j (n_j + \kappa)^{-\gamma-1}]$, and the difference of two numbers drawn from an interval is bounded by its length. Monotonicity of $(n + \kappa)^{-\gamma-1}$ in n gives the $n_{\min}$ form. $\square$

D IMPLEMENTATION AND REPRODUCIBILITY

Training schedule. The encoder is warm-started from the previous cycle and is never retrained from scratch. Each cycle uses a fresh optimizer and learning-rate schedule. All earlier real and generated images remain in the source set, so the setting differs from sequential-task learning in which previous data disappear. The cycle ablation holds the total optimizer update budget and the total number of generated images fixed and compares fewer repairs with more learning between them against more frequent repairs with less.

Core configuration. The main experiments use ResNet-50 encoders, five cycles of 100 epochs, mean k NN distance with $k = 100$, 500 anchors per cycle and five variants per anchor. The optimizer is AdamW with weight decay 10^{-4} , and method-specific learning rates and temperatures follow the settings of each baseline. The paired ViT-S experiments use the reference recipe of each objective. DINO uses two global and eight local crops at 96×96 with a bottleneck projection head, and MAE runs five cycles of 80 epochs.

Scoring and selection. Scoring uses the mean squared normalized ℓ_2 distance of Equation 4. For unit vectors the squared distance is an affine function of the cosine, so cosine scoring selects the identical anchor set. The mode-window selector restricts the score distribution to the 75th to 99th percentile range, builds a histogram with the automatic bin heuristic, forms a mode-centered candidate pool of size $4B$ and applies greedy farthest-point sampling on normalized features to select the final B anchors.

Generators. The main method uses Stable Diffusion 3 Medium in image-to-image mode, where an empty prompt makes the guidance scale inert. The generation ablation replaces it with FLUX or with the oracle, which adds the withheld real images and runs no generator. In the PASS and DiffusionDB experiments the SD3 batch size is 12 to fit the 24-hour wall-time limit.

Source data. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. The long-tailed ImageNet-100, CIFAR-10 and CIFAR-100 sources use their standard long-tailed configurations without the 10,000-image restriction. The ImageNet-100 long tail is built from the `clane9/imagenet-100` train and validation split. Class ranks 1 to 100 receive Pareto weights $r^{-1/\alpha}$ with $\alpha = 2.5$, the weights are min-max normalized and mapped linearly onto retention counts from 1,280 down to 5 images, the resulting per-class ratios are permuted uniformly at random over the classes, and each image is retained by an independent Bernoulli draw with the ratio of its class. The balanced source of Section 6.1 reads the same split and replaces the per-class ratios by their mean, so it removes the same expected fraction of images uniformly over classes.

Seeds and reporting. All claims rest on three-seed averages reported as mean and standard deviation. Paired comparisons share seeds and, where stated, source checkpoints. Multiple comparisons within one table are corrected by the Holm procedure. No two conditions within a table share an encoder fingerprint.

Code availability. An anonymous code release is available at <https://anonymous.4open.science/status/FOMO-0488>.

E COMPUTE RESOURCES AND RUNTIME

Most experiments use one node with two NVIDIA A100 80GB GPUs under a 24-hour wall-time limit. The DINO runs fit on a single 48GB card at the batch size used. The 454 completed runs retained for this paper total approximately 5,471 wall-clock hours, of which the baselines and ablations that appear only in the appendix account for 2,987. Runs on two-GPU nodes consume roughly twice that in GPU hours, while the paired ViT-S runs hold a single card each. Failed runs, timeouts, cache migration and pilot jobs are excluded from these totals.

Mean wall-clock times over three seeds are 16.9 hours for SimCLR and 21.8 for BRIDGE on ImageNet-100-LT, 10.2 and 14.9 on PASS, and 11.6 and 16.4 on DiffusionDB. The loop therefore raises wall-clock time by roughly 30 to 45 percent over the corresponding no-repair run, of which the embedding pass is a small fraction and diffusion sampling the remainder. Four repairs add 10,000 images, which is 10 percent of a 100,000-image source and 1 percent of a one-million-image source. Whether a fixed budget still helps at 100,000 or one million images was not tested.

F ASSETS AND TERMS OF USE

ImageNet-100-LT is derived from ImageNet. The ImageNet access terms restrict the images to non-commercial research and educational use. PASS is released under Creative Commons Attribution 4.0 and DiffusionDB under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers and Oxford-IIIT Pets are used from their official academic releases under their original terms. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation under the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third-party datasets or generated image corpora.

G BROADER IMPACT

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful where balanced data are difficult to collect. The method also relies on large image generators and web-sourced datasets, which raise risks of license violations, privacy and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms, release no new generators or scraped collections, and any deployment would need a review of these risks for its own task.

H FULL RESULTS

Table 9: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 $\pm$ 0.5	46.8 $\pm$ 0.4	13.7 $\pm$ 0.3	19.6 $\pm$ 0.2	35.9 $\pm$ 7.2	38.6 $\pm$ 1.5	50.4 $\pm$ 0.5	39.8 $\pm$ 1.5
	TS	72.5 $\pm$ 0.4	44.0 $\pm$ 0.5	12.5 $\pm$ 1.2	15.0 $\pm$ 0.1	35.7 $\pm$ 1.8	40.6 $\pm$ 2.2	56.7 $\pm$ 1.2	39.6 $\pm$ 1.1
	SDCLR	41.1 $\pm$ 1.7	18.0 $\pm$ 0.6	1.4 $\pm$ 0.3	2.3 $\pm$ 0.5	10.8 $\pm$ 1.7	10.4 $\pm$ 0.3	21.9 $\pm$ 0.6	15.1 $\pm$ 0.8
	BRIDGE (ours)	75.8 $\pm$ 0.2	49.3 $\pm$ 0.2	16.4 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.5 $\pm$ 0.7
	BRIDGE+TS (ours)	74.2 $\pm$ 0.5	45.6 $\pm$ 0.8	9.4 $\pm$ 0.5	15.0 $\pm$ 0.2	25.5 $\pm$ 2.5	33.1 $\pm$ 2.8	48.3 $\pm$ 0.4	35.9 $\pm$ 1.1
	BRIDGE+SDCLR (ours)	43.8 $\pm$ 1.7	18.4 $\pm$ 0.7	2.0 $\pm$ 0.3	2.5 $\pm$ 0.8	10.0 $\pm$ 0.8	12.2 $\pm$ 3.1	21.8 $\pm$ 0.5	15.8 $\pm$ 1.1
CIFAR-10-LT	SimCLR	67.8 $\pm$ 0.2	38.0 $\pm$ 1.0	6.5 $\pm$ 0.4	16.0 $\pm$ 0.9	19.2 $\pm$ 1.6	24.0 $\pm$ 1.3	32.0 $\pm$ 0.7	29.1 $\pm$ 0.9
	TS	68.5 $\pm$ 1.3	36.0 $\pm$ 1.7	4.3 $\pm$ 0.2	10.3 $\pm$ 0.1	13.1 $\pm$ 2.0	19.7 $\pm$ 2.3	31.5 $\pm$ 1.2	26.2 $\pm$ 1.3
	SDCLR	39.6 $\pm$ 1.1	15.6 $\pm$ 0.2	1.1 $\pm$ 0.1	2.4 $\pm$ 0.9	12.0 $\pm$ 1.6	9.1 $\pm$ 1.8	16.5 $\pm$ 1.6	13.8 $\pm$ 1.1
	BRIDGE (ours)	78.1 $\pm$ 0.3	49.8 $\pm$ 0.4	9.2 $\pm$ 0.8	17.7 $\pm$ 1.0	18.2 $\pm$ 1.1	27.4 $\pm$ 2.7	39.0 $\pm$ 0.6	34.2 $\pm$ 1.0
	BRIDGE+TS (ours)	78.6 $\pm$ 0.3	46.1 $\pm$ 0.3	7.5 $\pm$ 0.9	11.1 $\pm$ 0.7	14.5 $\pm$ 0.6	25.1 $\pm$ 2.7	39.6 $\pm$ 0.8	31.8 $\pm$ 0.9
	BRIDGE+SDCLR (ours)	40.9 $\pm$ 0.7	17.9 $\pm$ 0.5	1.4 $\pm$ 0.1	2.2 $\pm$ 0.6	11.0 $\pm$ 2.4	10.0 $\pm$ 2.7	21.0 $\pm$ 0.6	14.9 $\pm$ 1.1
CIFAR-100-LT	SimCLR	59.0 $\pm$ 1.5	31.1 $\pm$ 1.6	4.1 $\pm$ 0.3	11.2 $\pm$ 1.0	12.6 $\pm$ 3.4	20.1 $\pm$ 1.0	28.7 $\pm$ 1.3	23.8 $\pm$ 1.4
	TS	56.3 $\pm$ 0.2	28.0 $\pm$ 0.1	2.8 $\pm$ 0.6	8.5 $\pm$ 0.2	15.5 $\pm$ 2.5	16.2 $\pm$ 3.1	26.2 $\pm$ 1.8	21.9 $\pm$ 1.2
	SDCLR	37.1 $\pm$ 0.8	15.7 $\pm$ 0.2	1.9 $\pm$ 0.1	2.4 $\pm$ 0.7	9.4 $\pm$ 1.0	9.9 $\pm$ 1.7	16.2 $\pm$ 0.3	13.2 $\pm$ 0.7
	BRIDGE (ours)	74.2 $\pm$ 0.5	48.1 $\pm$ 0.4	8.3 $\pm$ 1.0	18.2 $\pm$ 0.7	24.4 $\pm$ 2.3	27.6 $\pm$ 0.7	39.8 $\pm$ 1.1	34.4 $\pm$ 1.0
	BRIDGE+TS (ours)	73.7 $\pm$ 0.1	47.3 $\pm$ 0.5	7.9 $\pm$ 0.6	14.1 $\pm$ 0.5	18.2 $\pm$ 3.2	28.4 $\pm$ 0.4	41.3 $\pm$ 0.3	33.0 $\pm$ 0.8
	BRIDGE+SDCLR (ours)	42.0 $\pm$ 1.5	16.9 $\pm$ 0.9	1.3 $\pm$ 0.1	2.2 $\pm$ 0.6	10.6 $\pm$ 0.8	8.3 $\pm$ 0.8	20.2 $\pm$ 0.9	14.5 $\pm$ 0.8
PASS-10k	SimCLR	72.3 $\pm$ 0.2	45.5 $\pm$ 0.5	11.0 $\pm$ 0.2	17.5 $\pm$ 1.4	33.3 $\pm$ 1.3	33.3 $\pm$ 0.9	44.7 $\pm$ 0.9	36.8 $\pm$ 0.8
	TS	71.5 $\pm$ 0.2	43.5 $\pm$ 0.4	10.2 $\pm$ 1.2	17.2 $\pm$ 0.2	33.1 $\pm$ 2.1	36.4 $\pm$ 1.8	49.4 $\pm$ 0.4	37.3 $\pm$ 0.9
	SDCLR	42.2 $\pm$ 1.1	17.7 $\pm$ 0.8	1.8 $\pm$ 0.3	2.2 $\pm$ 0.8	10.6 $\pm$ 1.3	8.2 $\pm$ 1.6	20.8 $\pm$ 0.4	14.8 $\pm$ 0.9
	BRIDGE (ours)	73.9 $\pm$ 0.3	47.1 $\pm$ 0.4	13.6 $\pm$ 0.7	20.5 $\pm$ 0.7	37.7 $\pm$ 4.0	37.1 $\pm$ 0.7	49.1 $\pm$ 0.5	39.9 $\pm$ 1.0
	BRIDGE+TS (ours)	68.9 $\pm$ 0.6	40.6 $\pm$ 0.7	12.1 $\pm$ 0.4	14.1 $\pm$ 0.5	34.6 $\pm$ 2.9	31.6 $\pm$ 1.5	52.4 $\pm$ 0.6	36.4 $\pm$ 1.0
	BRIDGE+SDCLR (ours)	44.0 $\pm$ 0.9	18.8 $\pm$ 0.9	1.4 $\pm$ 0.3	1.8 $\pm$ 0.9	12.6 $\pm$ 2.6	12.4 $\pm$ 1.3	21.2 $\pm$ 1.1	16.0 $\pm$ 1.1
DiffusionDB-10k	SimCLR	74.4 $\pm$ 0.5	48.6 $\pm$ 0.2	10.9 $\pm$ 0.7	17.6 $\pm$ 0.8	31.2 $\pm$ 2.4	36.1 $\pm$ 1.3	45.3 $\pm$ 0.5	37.7 $\pm$ 0.9
	TS	72.3 $\pm$ 0.3	44.4 $\pm$ 0.3	10.7 $\pm$ 0.6	15.4 $\pm$ 0.6	33.1 $\pm$ 1.1	39.9 $\pm$ 2.7	48.8 $\pm$ 0.2	37.8 $\pm$ 0.8
	SDCLR	42.4 $\pm$ 0.2	18.4 $\pm$ 0.2	1.1 $\pm$ 0.2	2.4 $\pm$ 0.1	13.0 $\pm$ 3.7	8.2 $\pm$ 1.4	21.0 $\pm$ 0.6	15.2 $\pm$ 0.9
	BRIDGE (ours)	75.9 $\pm$ 0.4	49.9 $\pm$ 0.5	12.8 $\pm$ 1.0	20.0 $\pm$ 0.4	35.7 $\pm$ 1.1	37.7 $\pm$ 0.8	49.1 $\pm$ 0.5	40.1 $\pm$ 0.7
	BRIDGE+TS (ours)	67.8 $\pm$ 0.6	39.1 $\pm$ 0.5	9.4 $\pm$ 1.2	12.4 $\pm$ 0.4	30.3 $\pm$ 3.0	33.7 $\pm$ 1.9	50.8 $\pm$ 0.6	34.8 $\pm$ 1.2
	BRIDGE+SDCLR (ours)	43.8 $\pm$ 0.9	18.5 $\pm$ 0.3	1.3 $\pm$ 0.4	2.5 $\pm$ 0.0	14.3 $\pm$ 1.6	11.4 $\pm$ 0.6	21.9 $\pm$ 0.6	16.2 $\pm$ 0.6

Table 10: Full k NN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 $\pm$ 0.4	37.3 $\pm$ 0.2	6.0 $\pm$ 0.7	11.2 $\pm$ 0.4	32.2 $\pm$ 0.7	23.6 $\pm$ 0.3	43.9 $\pm$ 0.9	31.8 $\pm$ 0.5
	TS	68.4 $\pm$ 0.3	36.5 $\pm$ 0.1	4.7 $\pm$ 0.2	8.3 $\pm$ 0.5	26.6 $\pm$ 1.0	23.6 $\pm$ 1.7	52.2 $\pm$ 0.5	31.5 $\pm$ 0.6
	SDCLR	32.5 $\pm$ 0.4	10.9 $\pm$ 0.2	1.0 $\pm$ 0.3	2.1 $\pm$ 0.1	10.7 $\pm$ 0.7	4.7 $\pm$ 0.2	11.2 $\pm$ 0.2	10.4 $\pm$ 0.3
	BRIDGE (ours)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	7.0 $\pm$ 0.2	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.4
	BRIDGE+TS (ours)	65.4 $\pm$ 0.2	33.6 $\pm$ 0.3	3.1 $\pm$ 0.1	7.1 $\pm$ 0.3	18.2 $\pm$ 0.2	15.6 $\pm$ 0.7	40.8 $\pm$ 0.5	26.3 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	36.3 $\pm$ 0.8	13.0 $\pm$ 0.5	1.1 $\pm$ 0.3	2.4 $\pm$ 0.2	9.7 $\pm$ 0.3	5.4 $\pm$ 0.4	13.9 $\pm$ 1.0	11.7 $\pm$ 0.5
CIFAR-10-LT	SimCLR	66.9 $\pm$ 0.4	33.4 $\pm$ 0.3	3.7 $\pm$ 0.4	9.8 $\pm$ 0.7	16.5 $\pm$ 0.8	13.9 $\pm$ 0.8	24.0 $\pm$ 0.8	24.0 $\pm$ 0.6
	TS	66.3 $\pm$ 0.6	27.8 $\pm$ 0.4	2.4 $\pm$ 0.6	6.4 $\pm$ 0.2	10.5 $\pm$ 0.7	11.1 $\pm$ 0.4	22.0 $\pm$ 0.9	20.9 $\pm$ 0.5
	SDCLR	30.1 $\pm$ 0.3	10.0 $\pm$ 0.3	1.7 $\pm$ 0.3	2.2 $\pm$ 0.2	10.1 $\pm$ 0.7	4.4 $\pm$ 0.4	9.8 $\pm$ 0.3	9.7 $\pm$ 0.4
	BRIDGE (ours)	74.8 $\pm$ 0.4	41.0 $\pm$ 0.4	4.8 $\pm$ 0.5	10.0 $\pm$ 0.2	24.2 $\pm$ 0.6	16.2 $\pm$ 0.7	29.7 $\pm$ 0.3	28.7 $\pm$ 0.4
	BRIDGE+TS (ours)	77.6 $\pm$ 0.7	35.3 $\pm$ 0.5	3.1 $\pm$ 0.3	6.6 $\pm$ 0.4	16.3 $\pm$ 0.6	12.5 $\pm$ 1.0	26.7 $\pm$ 0.1	25.5 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	34.4 $\pm$ 1.2	11.4 $\pm$ 0.6	1.2 $\pm$ 0.2	2.2 $\pm$ 0.2	8.8 $\pm$ 1.5	5.0 $\pm$ 0.2	11.7 $\pm$ 1.1	10.7 $\pm$ 0.7
CIFAR-100-LT	SimCLR	59.3 $\pm$ 0.8	30.1 $\pm$ 0.9	3.0 $\pm$ 0.3	7.4 $\pm$ 0.4	13.7 $\pm$ 0.2	10.8 $\pm$ 0.4	21.8 $\pm$ 1.1	20.9 $\pm$ 0.6
	TS	53.1 $\pm$ 0.6	24.6 $\pm$ 0.4	2.4 $\pm$ 0.4	5.5 $\pm$ 0.3	9.3 $\pm$ 0.4	8.3 $\pm$ 0.2	18.7 $\pm$ 1.0	17.4 $\pm$ 0.5
	SDCLR	25.7 $\pm$ 1.2	8.5 $\pm$ 0.6	0.9 $\pm$ 0.4	1.9 $\pm$ 0.2	7.3 $\pm$ 0.3	3.9 $\pm$ 0.3	8.3 $\pm$ 1.1	8.1 $\pm$ 0.6
	BRIDGE (ours)	69.3 $\pm$ 0.5	39.5 $\pm$ 0.1	4.8 $\pm$ 0.6	8.9 $\pm$ 0.1	21.9 $\pm$ 0.1	15.5 $\pm$ 0.1	29.1 $\pm$ 0.7	27.0 $\pm$ 0.3
	BRIDGE+TS (ours)	64.3 $\pm$ 0.4	36.6 $\pm$ 0.4	2.9 $\pm$ 0.3	6.3 $\pm$ 0.1	16.4 $\pm$ 0.5	12.3 $\pm$ 0.0	28.2 $\pm$ 0.6	23.9 $\pm$ 0.3
	BRIDGE+SDCLR (ours)	31.9 $\pm$ 0.4	10.4 $\pm$ 0.5	0.8 $\pm$ 0.1	2.1 $\pm$ 0.5	7.5 $\pm$ 1.6	4.5 $\pm$ 0.1	9.1 $\pm$ 0.7	9.5 $\pm$ 0.6
PASS-10k	SimCLR	67.2 $\pm$ 0.2	37.0 $\pm$ 0.3	5.4 $\pm$ 0.3	10.3 $\pm$ 0.5	30.4 $\pm$ 0.0	16.7 $\pm$ 0.5	38.3 $\pm$ 0.3	29.3 $\pm$ 0.3
	TS	65.2 $\pm$ 0.7	34.6 $\pm$ 0.3	4.4 $\pm$ 0.2	7.9 $\pm$ 0.2	24.5 $\pm$ 0.1	14.4 $\pm$ 0.4	39.3 $\pm$ 0.4	27.2 $\pm$ 0.3
	SDCLR	32.4 $\pm$ 1.5	11.5 $\pm$ 0.8	1.3 $\pm$ 0.2	2.0 $\pm$ 0.1	8.6 $\pm$ 0.7	5.2 $\pm$ 0.2	11.1 $\pm$ 0.8	10.3 $\pm$ 0.6
	BRIDGE (ours)	67.5 $\pm$ 0.3	36.9 $\pm$ 0.3	5.8 $\pm$ 0.5	10.7 $\pm$ 0.4	33.7 $\pm$ 0.2	17.8 $\pm$ 0.6	41.4 $\pm$ 0.8	30.6 $\pm$ 0.4
	BRIDGE+TS (ours)	58.8 $\pm$ 0.5	27.9 $\pm$ 0.5	3.0 $\pm$ 0.2	6.6 $\pm$ 0.1	23.0 $\pm$ 0.3	13.4 $\pm$ 0.4	38.8 $\pm$ 0.8	24.5 $\pm$ 0.4
	BRIDGE+SDCLR (ours)	37.1 $\pm$ 2.0	13.6 $\pm$ 1.0	1.6 $\pm$ 0.2	2.4 $\pm$ 0.2	10.1 $\pm$ 0.5	6.3 $\pm$ 0.3	13.1 $\pm$ 0.6	12.0 $\pm$ 0.7
DiffusionDB-10k	SimCLR	67.3 $\pm$ 0.1	36.2 $\pm$ 0.4	4.3 $\pm$ 0.2	8.8 $\pm$ 0.1	26.8 $\pm$ 0.4	17.3 $\pm$ 0.1	36.7 $\pm$ 0.8	28.2 $\pm$ 0.3
	TS	64.7 $\pm$ 0.4	33.6 $\pm$ 0.3	3.1 $\pm$ 0.5	7.3 $\pm$ 0.3	23.2 $\pm$ 0.4	17.0 $\pm$ 0.0	37.8 $\pm$ 0.6	26.7 $\pm$ 0.4
	SDCLR	32.6 $\pm$ 1.2	10.0 $\pm$ 0.3	1.3 $\pm$ 0.1	2.2 $\pm$ 0.2	6.1 $\pm$ 0.7	4.8 $\pm$ 0.4	10.6 $\pm$ 0.5	9.7 $\pm$ 0.5
	BRIDGE (ours)	68.3 $\pm$ 0.2	37.4 $\pm$ 0.4	5.4 $\pm$ 0.3	9.8 $\pm$ 0.1	30.4 $\pm$ 0.2	18.6 $\pm$ 0.2	39.4 $\pm$ 1.0	29.9 $\pm$ 0.3
	BRIDGE+TS (ours)	57.5 $\pm$ 0.8	26.6 $\pm$ 0.3	3.0 $\pm$ 0.4	6.5 $\pm$ 0.1	21.7 $\pm$ 0.8	15.5 $\pm$ 0.4	37.0 $\pm$ 0.4	24.0 $\pm$ 0.5
	BRIDGE+SDCLR (ours)	35.2 $\pm$ 0.7	11.1 $\pm$ 0.3	1.0 $\pm$ 0.1	2.2 $\pm$ 0.1	8.1 $\pm$ 1.0	5.0 $\pm$ 0.3	11.3 $\pm$ 0.7	10.6 $\pm$ 0.5

Table 11: Augmentation mechanism ablation with full linear-probe results. The oracle row restores the withheld real images.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	75.0 $\pm$ 0.3	48.5 $\pm$ 0.2	16.3 $\pm$ 0.8	20.7 $\pm$ 0.4	35.7 $\pm$ 0.9	41.8 $\pm$ 0.8	52.4 $\pm$ 0.7	41.5 $\pm$ 0.6
FLUX	70.7 $\pm$ 0.4	42.1 $\pm$ 1.4	10.3 $\pm$ 1.3	17.5 $\pm$ 0.9	29.4 $\pm$ 4.6	31.5 $\pm$ 2.9	43.7 $\pm$ 1.4	35.0 $\pm$ 1.9
Oracle (withheld real images restored)	74.8 $\pm$ 0.4	47.8 $\pm$ 0.7	13.1 $\pm$ 0.1	20.5 $\pm$ 0.5	40.3 $\pm$ 6.1	38.1 $\pm$ 1.5	50.3 $\pm$ 0.4	40.7 $\pm$ 1.4

Table 12: Augmentation mechanism ablation with full k NN results. The oracle row restores the withheld real images.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3 (default)	68.6 $\pm$ 0.1	37.2 $\pm$ 0.2	6.4 $\pm$ 0.6	11.5 $\pm$ 0.2	35.2 $\pm$ 0.3	24.3 $\pm$ 0.6	46.8 $\pm$ 0.7	32.9 $\pm$ 0.4
FLUX	65.8 $\pm$ 1.0	35.4 $\pm$ 0.8	5.7 $\pm$ 0.7	9.4 $\pm$ 0.9	27.7 $\pm$ 1.0	18.8 $\pm$ 1.3	38.7 $\pm$ 1.2	28.8 $\pm$ 1.0
Oracle (withheld real images restored)	68.8 $\pm$ 0.2	37.7 $\pm$ 0.3	6.6 $\pm$ 0.5	11.4 $\pm$ 0.5	32.7 $\pm$ 0.1	23.6 $\pm$ 0.6	44.4 $\pm$ 0.4	32.2 $\pm$ 0.4

Table 13: Selection rule ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	75.6 $\pm$ 0.1	49.2 $\pm$ 0.2	16.9 $\pm$ 0.7	21.9 $\pm$ 0.4	34.6 $\pm$ 2.9	43.5 $\pm$ 0.7	52.8 $\pm$ 0.1	42.1 $\pm$ 0.7
Mode-window q50-99 + diversity	75.5 $\pm$ 0.1	49.3 $\pm$ 0.4	15.2 $\pm$ 0.3	22.4 $\pm$ 0.6	35.5 $\pm$ 1.1	43.8 $\pm$ 3.1	53.4 $\pm$ 0.9	42.2 $\pm$ 0.9
Mode-window q75-99 + diversity (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
Top-tail	70.8 $\pm$ 0.5	43.1 $\pm$ 1.0	10.0 $\pm$ 1.7	15.9 $\pm$ 1.3	27.7 $\pm$ 1.0	30.5 $\pm$ 1.2	42.2 $\pm$ 1.4	34.3 $\pm$ 1.2
Uniform	71.5 $\pm$ 0.2	43.8 $\pm$ 0.7	9.3 $\pm$ 0.9	17.5 $\pm$ 1.2	32.9 $\pm$ 0.4	31.7 $\pm$ 3.9	42.0 $\pm$ 0.9	35.5 $\pm$ 1.2

Table 14: Selection rule ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window q1-99 + diversity	69.2 $\pm$ 0.3	37.7 $\pm$ 0.2	6.6 $\pm$ 0.2	11.4 $\pm$ 0.2	35.9 $\pm$ 0.2	25.0 $\pm$ 0.5	48.3 $\pm$ 0.3	33.4 $\pm$ 0.3
Mode-window q50-99 + diversity	68.7 $\pm$ 0.1	38.0 $\pm$ 0.2	7.3 $\pm$ 0.2	11.3 $\pm$ 0.2	36.5 $\pm$ 0.1	24.8 $\pm$ 0.8	48.5 $\pm$ 0.4	33.6 $\pm$ 0.3
Mode-window q75-99 + diversity (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
Top-tail	65.7 $\pm$ 0.8	34.4 $\pm$ 0.2	4.1 $\pm$ 0.4	9.6 $\pm$ 0.2	23.9 $\pm$ 0.9	18.1 $\pm$ 0.8	35.4 $\pm$ 1.2	27.3 $\pm$ 0.6
Uniform	66.1 $\pm$ 1.0	35.7 $\pm$ 0.3	4.9 $\pm$ 0.1	10.2 $\pm$ 0.7	25.3 $\pm$ 0.4	19.0 $\pm$ 0.8	37.4 $\pm$ 0.6	28.4 $\pm$ 0.6

Table 15: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	68.9 $\pm$ 0.6	41.6 $\pm$ 1.3	10.6 $\pm$ 1.2	16.4 $\pm$ 0.4	30.7 $\pm$ 4.5	33.2 $\pm$ 0.9	43.3 $\pm$ 0.6	35.0 $\pm$ 1.4
5 cycles (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
10 cycles	70.5 $\pm$ 1.2	43.8 $\pm$ 0.6	9.2 $\pm$ 0.9	17.2 $\pm$ 1.6	24.8 $\pm$ 5.2	31.3 $\pm$ 1.7	41.9 $\pm$ 0.6	34.1 $\pm$ 1.7

Table 19: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both runs of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean $\pm$ standard deviation over three seeds. Linear-probe accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	7.15 $\pm$ 0.19	10.92 $\pm$ 0.40	32.90 $\pm$ 2.72	25.00 $\pm$ 1.25	60.46 $\pm$ 0.46	34.96 $\pm$ 0.20	41.95 $\pm$ 1.43
SimCLR, with BRIDGE	8.18 $\pm$ 0.42	11.82 $\pm$ 0.43	36.60 $\pm$ 2.85	29.80 $\pm$ 2.98	60.85 $\pm$ 0.35	36.33 $\pm$ 0.14	43.93 $\pm$ 1.09
MoCo v3, SSL baseline	7.57 $\pm$ 0.23	11.05 $\pm$ 1.31	30.28 $\pm$ 1.36	29.26 $\pm$ 3.13	59.96 $\pm$ 0.39	34.47 $\pm$ 0.01	44.79 $\pm$ 1.17
MoCo v3, with BRIDGE	9.55 $\pm$ 0.34	13.42 $\pm$ 0.98	36.82 $\pm$ 6.07	31.25 $\pm$ 4.00	62.36 $\pm$ 0.46	36.30 $\pm$ 0.13	47.55 $\pm$ 0.86
DINO, SSL baseline	8.10 $\pm$ 0.25	11.40 $\pm$ 1.03	38.13 $\pm$ 5.24	27.54 $\pm$ 2.34	58.93 $\pm$ 0.94	33.56 $\pm$ 1.75	44.63 $\pm$ 1.47
DINO, with BRIDGE	9.49 $\pm$ 0.88	13.28 $\pm$ 0.58	37.25 $\pm$ 2.61	34.60 $\pm$ 3.62	61.30 $\pm$ 0.07	35.23 $\pm$ 0.31	46.78 $\pm$ 0.77
MAE, SSL baseline	1.54 $\pm$ 0.19	2.44 $\pm$ 1.00	10.46 $\pm$ 5.88	10.05 $\pm$ 3.03	37.69 $\pm$ 3.73	13.20 $\pm$ 1.94	19.93 $\pm$ 2.13
MAE, with BRIDGE	1.56 $\pm$ 0.43	3.18 $\pm$ 0.70	8.50 $\pm$ 2.26	8.61 $\pm$ 2.20	41.46 $\pm$ 2.20	15.00 $\pm$ 2.40	22.19 $\pm$ 2.35

Table 20: Paired full-schedule ViT-S transfer on ImageNet-100-LT, with every downstream dataset reported separately. Both runs of each pair start from the same source checkpoint and receive the same number of post-branch optimizer updates. Four objectives span contrastive (SimCLR, MoCo v3), self-distillation (DINO) and masked-reconstruction (MAE) pretraining. The first three run five cycles of 100 epochs and MAE five of 80. Mean $\pm$ standard deviation over three seeds. k NN accuracy.

Method	Cars	Aircraft	Flowers	Pets	CIFAR-10	CIFAR-100	IN100-LT
SimCLR, SSL baseline	6.26 $\pm$ 0.17	7.09 $\pm$ 0.17	36.53 $\pm$ 0.89	19.16 $\pm$ 0.64	66.05 $\pm$ 0.48	36.72 $\pm$ 0.36	41.21 $\pm$ 0.72
SimCLR, with BRIDGE	6.52 $\pm$ 0.22	7.56 $\pm$ 0.23	38.29 $\pm$ 0.83	19.93 $\pm$ 0.53	65.70 $\pm$ 0.85	36.46 $\pm$ 0.33	42.91 $\pm$ 0.78
MoCo v3, SSL baseline	6.65 $\pm$ 0.38	7.71 $\pm$ 0.16	38.91 $\pm$ 1.03	23.56 $\pm$ 1.12	69.29 $\pm$ 0.25	39.98 $\pm$ 0.44	45.14 $\pm$ 0.46
MoCo v3, with BRIDGE	6.74 $\pm$ 0.26	8.21 $\pm$ 0.18	40.51 $\pm$ 1.45	24.91 $\pm$ 0.45	65.41 $\pm$ 0.21	36.24 $\pm$ 0.46	47.71 $\pm$ 1.53
DINO, SSL baseline	6.05 $\pm$ 0.27	7.73 $\pm$ 0.52	38.52 $\pm$ 1.19	19.26 $\pm$ 0.16	62.69 $\pm$ 2.42	33.98 $\pm$ 2.80	45.28 $\pm$ 0.98
DINO, with BRIDGE	6.39 $\pm$ 0.22	8.16 $\pm$ 0.26	40.72 $\pm$ 0.31	21.79 $\pm$ 1.15	64.38 $\pm$ 3.51	35.59 $\pm$ 2.51	47.14 $\pm$ 0.29
MAE, SSL baseline	1.00 $\pm$ 0.14	2.24 $\pm$ 0.29	5.30 $\pm$ 0.52	4.58 $\pm$ 0.17	31.34 $\pm$ 1.65	10.96 $\pm$ 0.44	11.75 $\pm$ 0.38
MAE, with BRIDGE	1.38 $\pm$ 0.37	2.50 $\pm$ 0.70	7.54 $\pm$ 2.13	5.35 $\pm$ 0.76	35.58 $\pm$ 7.14	13.44 $\pm$ 3.46	15.46 $\pm$ 2.68

Table 16: Cycle count ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.3 $\pm$ 0.4	34.2 $\pm$ 0.5	5.0 $\pm$ 0.7	10.0 $\pm$ 0.2	26.1 $\pm$ 0.8	18.7 $\pm$ 0.6	35.9 $\pm$ 0.3	27.9 $\pm$ 0.5
5 cycles (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
10 cycles	66.4 $\pm$ 0.5	35.4 $\pm$ 0.3	4.8 $\pm$ 0.3	10.4 $\pm$ 0.6	25.2 $\pm$ 0.8	18.6 $\pm$ 0.9	36.0 $\pm$ 0.7	28.1 $\pm$ 0.6

Table 17: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	68.8 $\pm$ 0.1	41.7 $\pm$ 0.2	13.5 $\pm$ 0.3	19.7 $\pm$ 0.6	38.7 $\pm$ 3.4	38.4 $\pm$ 2.6	50.2 $\pm$ 0.2	38.7 $\pm$ 1.1
ResNet-50 (default)	76.0 $\pm$ 0.3	49.1 $\pm$ 0.2	15.2 $\pm$ 1.2	22.4 $\pm$ 0.4	36.8 $\pm$ 1.9	42.9 $\pm$ 1.0	53.8 $\pm$ 0.1	42.3 $\pm$ 0.7
ViT-S	62.7 $\pm$ 0.4	37.6 $\pm$ 0.6	8.8 $\pm$ 0.5	12.2 $\pm$ 0.2	35.1 $\pm$ 4.2	30.2 $\pm$ 1.6	42.5 $\pm$ 1.2	32.7 $\pm$ 1.3
ViT-B	66.0 $\pm$ 2.3	40.6 $\pm$ 2.3	8.3 $\pm$ 0.5	14.3 $\pm$ 0.8	38.8 $\pm$ 2.6	31.3 $\pm$ 2.1	44.5 $\pm$ 1.7	34.8 $\pm$ 1.8

Table 18: Backbone architecture ablation with full k NN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	66.3 $\pm$ 0.2	35.7 $\pm$ 0.3	5.9 $\pm$ 0.2	11.1 $\pm$ 0.4	33.5 $\pm$ 0.1	23.3 $\pm$ 0.5	44.5 $\pm$ 0.7	31.5 $\pm$ 0.4
ResNet-50 (default)	69.0 $\pm$ 0.5	37.8 $\pm$ 0.3	6.7 $\pm$ 0.3	11.7 $\pm$ 0.2	36.5 $\pm$ 0.4	25.3 $\pm$ 1.0	48.1 $\pm$ 0.6	33.6 $\pm$ 0.5
ViT-S	67.3 $\pm$ 0.4	37.5 $\pm$ 0.3	5.3 $\pm$ 0.2	7.6 $\pm$ 0.1	36.2 $\pm$ 0.7	20.1 $\pm$ 1.0	42.1 $\pm$ 1.0	30.9 $\pm$ 0.5
ViT-B	69.3 $\pm$ 0.8	40.1 $\pm$ 0.8	6.0 $\pm$ 0.2	8.3 $\pm$ 0.2	39.8 $\pm$ 0.8	21.9 $\pm$ 1.4	45.1 $\pm$ 0.7	32.9 $\pm$ 0.7

I ALGORITHM

Algorithm 1 BRIDGE training loop with mode-window selection

```

1: Initialize source set  $D^{(0)}$ , encoder  $f_{\theta^{(0)}}$ , frozen generator  $g_\phi$ , cycle lengths  $\{E_c\}_{c=0}^{C-1}$ 
2: for  $c = 0, \dots, C - 1$  do
3:   Continue training  $f_{\theta^{(c)}}$  on  $D^{(c)}$  for  $E_c$  epochs with a fresh optimizer schedule
4:   if  $c < C - 1$  then
5:     Compute normalized embeddings  $Z^{(c)} = \{f_{\theta^{(c)}}(x_i) \mid x_i \in D^{(c)}\}$ 
6:     Compute mean  $k$ NN scores  $\{d_i^{(c)}\}$  from  $Z^{(c)}$ 
7:     Restrict to the 75th to 99th percentile band and locate the modal bin center  $\bar{d}^{(c)}$ 
8:     Form the candidate pool  $P^{(c)}$  of the  $\alpha B$  images with scores closest to  $\bar{d}^{(c)}$ 
9:     Select  $S^{(c)} \subseteq P^{(c)}$  with  $|S^{(c)}| = B$  by greedy farthest-point sampling in embedding space
10:    Form  $\tilde{D}^{(c)} = \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)}, j \in [G]\}$  and set  $D^{(c+1)} = D^{(c)} \cup \tilde{D}^{(c)}$ 
11:   end if
12: end for
13: Return the final encoder  $f_{\theta^{(C-1)}}$ 

```
